MIL-V-0050067D(MU)
22 January 1970
USED IN LIEU OF
MIL-V-50067C
30 September 1964

MILITARY SPECIFICATION

VOICEMITTER-OUTLET VALVE, C29R1

COMPONENTS FOR

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-V-50067C. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers the components for the C29R1 voicemitter-outlet valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-81-3	-	Voicemitter-Outlet Valve, C29R1 - Data List.
A18-60-761	-	Outlet Valve Seat.
DL136-13-342	-	Impact and Rigidity Mechanical Tester, Q26.
E136-14-3-1	-	Speech Diaphragm Pressure Test Device, Q25.
D136-45-202-15	-	Outlet Valve Leakage Indicator, M4A1.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of this publication should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing DL5-81-3 or on the subsidiary drawings, shall conform to the specifications, thereon or to the specific characteristics set forth on the drawings.

3.2 Manufacture and assembly. The components shall be manufactured and assembled as specified on the drawings which are listed on Drawing DL5-81-3.

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3.3 Valve seat leakage. Leakage through the valve seat shall not exceed 15 milliliters of air per minute at a pressure differential of 25 millimeters of water when tested as specified in 4.4.4.1.

3.4 Frame-retaining ring and outlet-valve seat assembly. The crimped joint of the frame-retaining ring and the cemented joint between the frame and outlet-valve seat assembly shall not leak when tested as specified in 4.4.4.2. The preproduction sample shall not leak when tested as specified in 4.4.4.3.

3.5 Impact resistance. The outlet valve seat shall not break nor shall the ribs or hub be depressed so that the distance from any portion of the upper surface of the hub to the upper edge of the sealing-rim surface (see Dwg A18-60-761) has been increased by more than 0.012 inch, when tested as specified in 4.4.4.4.

3.6 Preproduction sample. Prior to the start of regular production, a preproduction sample of each component, except those components covered by separate specifications, shall be produced, in accordance with this specification, for examination and tests (see 4.3).

3.7 Workmanship. The voicemitter-outlet valve components shall be free from imperfections, foreign matter, and damage. Imperfections shall include blisters, porosity, cracks, tears, and excessive flash for the rubber components, and dents, cracks, burrs, sharp edges, surface roughness, and distortion for the metal components. Damage shall constitute tears, cracks, dents, and distortion as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 5 of each component shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The preproduction sample shall be examined for all requirements of the drawings and of this specification.

4.3.2.2 For tests. The preproduction sample shall be tested in accordance with 4.4.4.

4.3.3 Tests. Tests shall be conducted as specified in 4.4.4.

4.3.4 Acceptance/rejection criteria. The preproduction sample of each component shall meet the examination and tests specified by 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of like components produced by one manufacturer from the same materials, at the same plant, and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105 using level S-4.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample components shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

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4.4.3.2 For destructive tests. Sample components shall be tested in accordance with 4.4.4.4 using an AQL of 0.65 percent defective.

4.4.3.3 Classification of defects.

(a) Ring, sealing (Dwg B5-81-4).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (2.425)
102	Depth (0.235)
*103	Radii
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Diameter (2.475)
*202	Depth (0.105)
*203	Gap (5/64)
204	Protective finish incorrect or missing
*205	Diameter (2.015)
*206	Diameter (2.490)

*Tool controlled (may be periodically inspected).

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4.4.3.3 Classification of defects (continued).

(b) Ring, retaining (Dwg B5-81-5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (2.417)
102	Diameter (2.465)
103	Diameter (2.367)
104	Depth (0.178)
105	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Diameter (2.064)
*202	Depth (0.354)
*203	Height (0.095)
204	Protective finish missing or incorrect

*Tool controlled (may be periodically inspected).

(c) Gasket (Dwg B5-81-6).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (3.7)
*102	Outside diameter
103	Thickness
104	Foreign matter (grease or oil)

*Tool controlled (may be periodically inspected).

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4.4.3.3 Classification of defects (continued).

(d) Frame (Dwg C5-81-7).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Plain view profile incorrect
102	Diameter (1.498)
103	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Diameter (1.515)
*202	Diameter (1.255)
*203	Depth (11/16)
*204	Diameter (2.070)
205	Protective finish missing or incorrect

*Tool controlled (may be periodically inspected).

(e) Baffle, front (Dwg C5-81-8).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter
102	Depth (0.060)
103	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Cutouts incorrect or incorrectly located
202	Protective finish incorrect or missing

*Tool controlled (may be periodically inspected).

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4.4.3.3 Classification of defects (continued).

(f) Baffle, cupped (Dwg C5-81-9).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (2.315)
*102	Diameter (2.130)
103	Flange height
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
201	Depth (0.145)
202	Cutouts incorrect or incorrectly located

*Tool controlled (may be periodically inspected).

(g) Cup, inner (Dwg B5-81-10).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
*101	Outside diameter
*102	Inside diameter
103	Flange height
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
201	Radius

*Tool controlled (may be periodically inspected).

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4.4.3.3 Classification of defects (continued).

(h) Baffle, inner (Dwg C5-81-11).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter
102	Projection depth (0.062) or radius
103	Depth (0.020)
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Diameter (2.031)
*202	Location of Projections incorrect
203	Cutouts incorrect or incorrectly located
204	Protective finish incorrect or missing

*Tool controlled (may be periodically inspected).

(i) Gasket, baffle (Dwg B5-81-12).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (3.7)
*102	Outside diameter
*103	Inside diameter
104	Thickness
105	Foreign matter (grease or oil)

*Tool controlled (may be periodically inspected).

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4.4.3.3 Classification of defects (continued).

(j) Frame and retaining-ring assembly (Dwg C5-81-14).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.25 percent defective	
101	Valve seat leakage	4.4.4.1
102	Frame-retaining ring and outlet valve seat assembly leakage	4.4.4.2
103	Component missing or incorrectly assembled	
104	Frame-retaining ring crimped joint defective or incorrect	
105	Workmanship (3.7)	
Minor:	AQL 2.5 percent defective	
201	Radii or retaining beads missing	
202	Protective finish missing or incorrect	

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4.4.3.3 Classification of defects (continued).

(k) Seat, outlet valve (Dwg C5-81-16).

Categories	Defects	Acceptance standard
Critical:	None defined	
Major:	AQL 0.25 percent defective	
*101	Valve seat leakage	4.4.4.1
	AQL 1.0 percent defective	
102	Workmanship (3.7)	
103	Diameter (1.130)	
104	Diameter (1.512)	
105	Diameter (0.176)	
106	Thickness (0.150)	
107	Thickness (0.160)	
108	Sealing rim surface not smooth	
109	Countersink diameter (0.195)	
Minor:	AQL 2.5 percent defective	
201	Marking incorrect, missing, or illegible	

*For prime procurement only.

4.4.4 Tests.

4.4.4.1 Valve seat leakage. An outlet valve disc (Dwg B5-2-1006) shall be assembled in the frame, in accordance with Drawing D5-81-3, and tested using the M4A1 Outlet Valve Leakage Indicator (D136-45-202-15) (see 6.3).

4.4.4.2 Frame-retaining ring and outlet-valve seat-assembly leakage. The leakage of the frame-retaining ring and outlet-valve seat assembly shall be determined using the Q25 Speech Diaphragm Pressure Test Device (E136-14-3-1), at a static pressure of 3 pounds per square inch, for 15 seconds.

4.4.4.3 Preproduction sample. Repeat the leakage test specified in 4.4.4.2 on the preproduction sample after the assemblies have been subjected to a temperature of -40° ± 2° F for 24 hours and again after they have been subjected to boiling water for two hours.

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4.4.4.4 Impact resistance. Impact resistance of the outlet valve seat shall be determined using the Q26 Impact and Rigidity Mechanical Tester (DL136-13-342), with a 1/4-pound weight dropped from a height of 24 inches.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. Components of like description shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Like components, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing D150-2-1.

5.4 Repair part. When these items are procured for storage and issue as repair parts, preservation, packaging, packing, and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. The components covered by this specification are intended to be used in the manufacture of the C29R1 voicemitter-outlet valve.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Which component.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation before award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not sample is acceptable.

6.3 Outlet valve disc. The outlet valve discs required for use in performing the valve seat leakage test will be furnished by the Government and should be obtained as directed by the contracting officer.

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6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 21, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:

Army - MU
Navy - SH
Air Force - 84

Preparing activity:

Army - MU(EA)

Project No. 4240-A380

Review activities:

Army - MD
Navy - SH

User activities:

Navy - MC
Air Force - 84

International interest (see 6.4)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER(S)

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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